Pneumatic Reservoir Computing in a Fabric Soft Robotic Arm:

Sensing Topology, Stiffness-Modulating Pre-Inflation

and Control

by

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ABSTRACT

Physical reservoir computing (PRC) can exploit the dynamics of soft robots as a computational substrate, but performance depends strongly on how internal mechanical state is sensed. This work studies a fabric-based pneumatic soft arm with a five-pouch sensing segment under two extreme pneumatic sensing topologies: *parallel* (independently sealed pouches) and *coupled* (pouches connected by airflow).

Across 36 trials spanning excitation waveform (axial, circular, triangular), sensing-segment pre-inflation (1–3 PSI), and actuation pressure range ($P_{\max} \in \{5, 10\}$ PSI), the parallel topology yields a higher-dimensional pressure state (lower inter-pouch correlation, higher delay-decoding index) and improves 1-s-ahead bending-angle prediction with a linear ridge-regression readout when all five sensing pressures are available (median NMSE 0.242 vs 0.774). The coupled topology, however, increases pressure–angle sensitivity (0.0071 vs 0.0044 PSI/deg), indicating stronger direct coupling between deformation and the internal pneumatic state.

Finally, a sensor-ablation study shows that, in the parallel topology, prediction accuracy improves rapidly as additional independent pouches are added: the best fixed two-pouch subset achieves median NMSE 0.344 and the best fixed three-pouch subset achieves median NMSE 0.291, compared to 0.242 using all five pouches. When allowing oracle per-trial subset selection, three pouches achieve a median NMSE of 0.234, indicating that redundant channels can sometimes degrade generalization. Augmenting PRC with commanded-pressure context (PRC+context) further reduces prediction error to a median NMSE of 0.056, yielding a context-conditioned observer suitable for model-based prediction and future feedback control.

ACKNOWLEDGMENTS

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Chapter 1

SYSTEM DESIGN AND AUTOMATED DATA COLLECTION

System overview

This thesis investigates a pneumatically actuated, fabric-based soft robotic arm and its use as a physical dynamical system for data-driven modeling and physical reservoir computing (PRC). The experimental platform comprises:

- a four-segment pneumatic arm (Segments 1–4),
- a pneumatic drive system that regulates pressure applied to each segment,
- embedded sensing (pressure) and an external motion-capture (mocap) system,
- a networked embedded-control layer (microcontroller nodes) and a host computer that orchestrates experiments and logs synchronized data.

Figure 1.1 summarizes the segment and pouch nomenclature used throughout the report.

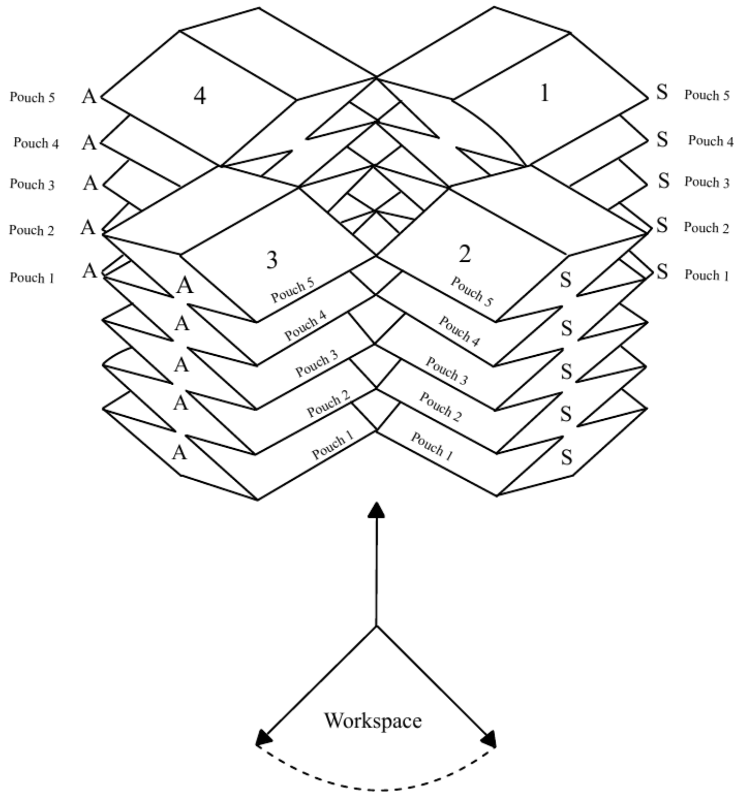

Figure 1.1: Arm segmentation and pouch numbering convention used in this work (Segment 1 proximal to Segment 4 distal; pouches numbered 1–5 within a segment).

Pneumatic arm and sensing layout

The arm is composed of four pneumatic segments. Two pressure-instrumentation patterns are used in the platform:

- **Pouch-resolved sensing (Segment 1):** individual pressure taps are placed on each of five pouches, yielding a five-dimensional pressure readout that serves as the PRC "reservoir state" in Chapter 2.

- **Segment-level monitoring (Segments 2–4):** a single pressure tap is placed on the supply/manifold line per segment, yielding one pressure measurement per segment for actuation tracking and diagnostics.

These two instrumented tubing layouts are shown in Figures 1.2 and 1.3. The diagrams indicate where pressure sensors tap into the pneumatic volume and where the pressure supply interfaces with the segment.

Role of Segment 1 in the PRC study. Chapter 2 focuses on physical reservoir computing using the five pouch pressures of **Segment 1** as the reservoir readout vector $\mathbf{s}(t)$. Segments 2–4 are commanded with time-varying pressure setpoints to excite the arm dynamics. Additional pressure channels (segment-level sensors for Segments 2–4 and raw ADC channels that are uninstrumented in the logged dataset) are recorded for diagnostics and post hoc analysis but are not used as PRC state inputs unless explicitly stated.

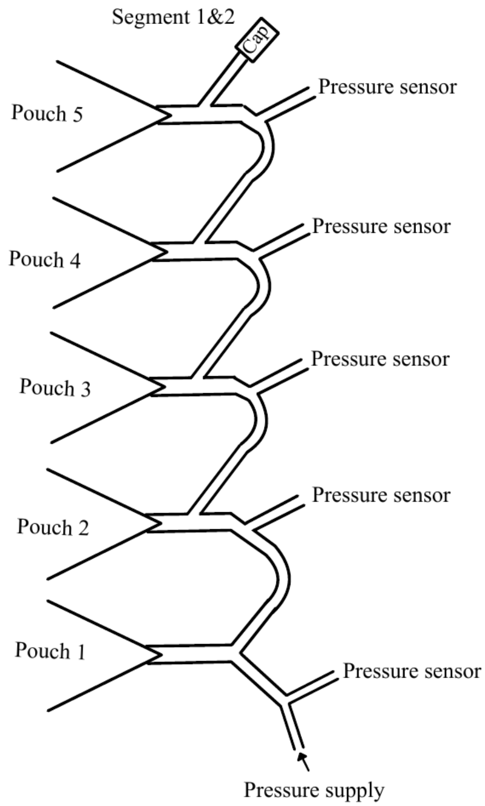

Figure 1.2: Instrumented plumbing layout for a five-pouch segment with per-pouch pressure taps. Segment 1 uses this pouch-resolved instrumentation for PRC; Segment 2 is monitored at segment level in the dataset analyzed here.

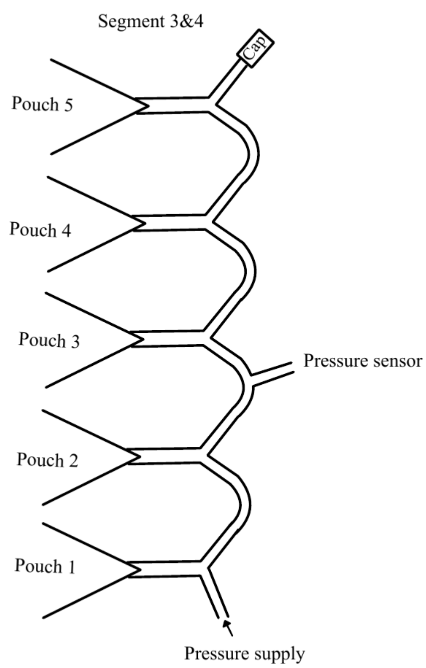

Figure 1.3: Instrumented plumbing layout for the segment-level monitored segments (Segments 3–4): a single pressure sensor tap on the supply/manifold line per segment.

Pneumatic leakage diagnosis and mitigation

Pneumatic leakage directly impacts repeatability and data quality in this platform because pressure drift changes the internal state of the arm even when commanded setpoints remain constant. During commissioning, leakage was observed in the pneumatic circuit and was identified to originate predominantly from the *tubing network* (rather than the fabric pouches themselves).

To mitigate this issue, inline **one-way (check) valves** were added to the supply tubing. These valves enforce directional flow from the regulator/supply toward the arm and reduce backflow and depressurization through upstream leak paths. In practice, adding one-way valves resolved most of the leakage issues observed during early testing. The automated sequence logic described in Section 1 also supports periodic refill/conditioning steps to help maintain consistent initial conditions.

Figure 1.4 shows a simplified schematic of the mitigation concept and where a check valve is inserted relative to the supply/regulation stage and the flexible tubing run.

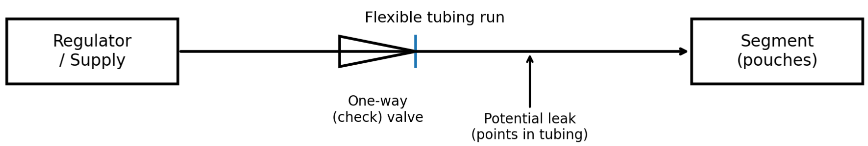

Figure 1.4: Conceptual schematic showing one-way (check) valve insertion on a segment supply line to mitigate tubing leakage and backflow. The diagram is illustrative of the implemented mitigation: a check valve placed inline between the regulation stage and the flexible tubing feeding a segment/pouch volume.

Embedded electronics and pressure I/O

Distributed embedded nodes

The pneumatic system is interfaced through four Ethernet-connected microcontroller nodes (Arduino IDs {3,6,7,8}), each corresponding to one arm segment for pressure command. At the host interface level, each node implements:

- **Command input:** a single 32-bit floating-point pressure setpoint (PSI) sent from the host via a TCP stream.
- **Sensor output:** four signed 16-bit integers representing raw ADC counts acquired from an external 16-bit ADC.

This interface allows the host to command one desired pressure per segment at the experiment rate, while simultaneously recording up to four analog sensor channels per node. In the February 2026 dataset used throughout this report, only a subset of the 16 logged analog channels carry meaningful pressure measurements (the remaining channels stay near zero and are removed during the cleaning stage).

Analog output path (pressure command)

Each embedded node generates an analog command voltage using a 12-bit $\mathrm{I^2C}$ DAC module (MCP4725). The DAC output is scaled from the received desired pressure setpoint and drives an analog-controlled pressure regulation stage (the regulator itself implements the low-level pressure regulation; the host operates at the level of pressure setpoints).

Analog input path (pressure sensing)

Each embedded node samples up to four analog pressure sensors using a 16-bit $\mathrm{I^2C}$ ADC module (ADS1115) configured with gain setting `GAIN_TWOTHIRDS` (full-scale $\pm 6.144\,\mathrm{V}$). The embedded firmware transmits the four raw ADC codes to the host as big-endian `int16` values.

On the host, raw counts are converted to voltage using the ADS1115 scaling

$$V = \frac{6.144}{32768}\, c, \tag{1.1}$$

where c is the signed 16-bit ADC count.

The voltage is then mapped to pressure using the linear transfer implemented in the logging software, which assumes a ratiometric pressure sensor output of $0.5\,\mathrm{V}$ at 0 PSI and $4.5\,\mathrm{V}$ at 30 PSI:

$$P\ [\mathrm{PSI}] = \frac{V - 0.5}{4.0} \times 30. \tag{1.2}$$

Equations (1.1)–(1.2) describe the conversion used for all logged pressure channels in the dataset.

Why this embedded architecture is suitable

The combination of (i) Ethernet-connected embedded nodes, (ii) external precision ADC/DAC modules, and (iii) a host-orchestrated experiment loop supports the following technical requirements:

- **Channel count and modularity:** four distributed nodes provide one pressure command channel per segment and scale naturally with additional segments.
- **Precision for low-pressure operation:** the external 16-bit ADC provides finer quantization than typical microcontroller 10-bit ADCs, which is beneficial when operating in the 0–10 PSI range.
- **Deterministic host scheduling:** pressure commands and logging are executed at a fixed cadence (100 Hz in this system), enabling repeatable excitation and consistent time bases across trials.

Motion capture and synchronization

The arm motion is recorded using an OptiTrack motion-capture system with three rigid bodies mounted along the arm. Rigid-body poses are published on ROS 2 topics (`/vrpn_mocap/Rigid_Body_001/pose`, `/vrpn_mocap/Rigid_Body_002/pose`, `/vrpn_mocap/Rigid_Body_003/pose`) as `geometry_msgs/PoseStamped`. A lightweight bridge node republishes the three poses as a ZeroMQ stream on `tcp://127.0.0.1:3885`.

Each published mocap sample contains 21 floating-point values:

$$\big[\mathbf{p}_1,\, \mathbf{q}_1,\, \mathbf{p}_2,\, \mathbf{q}_2,\, \mathbf{p}_3,\, \mathbf{q}_3\big], \tag{1.3}$$

where $\mathbf{p}_i = (x_i, y_i, z_i)$ is position and $\mathbf{q}_i = (q_{x,i}, q_{y,i}, q_{z,i}, q_{w,i})$ is the unit quaternion of rigid body $i \in \{1, 2, 3\}$.

On the host computer, the mocap stream is received in a dedicated thread and the most recent sample is time-stamped and attached to each logged control sample. The controller and logger loops both run with drift-corrected scheduling at $\approx 100\,\mathrm{Hz}$, so pressure, sensor, and mocap data are aligned on a common experiment time base.

Automated data collection pipeline

Experiment orchestration

Experiments are executed by a host-side controller that runs three concurrent loops:

1. **Waveform generation and command dispatch:** computes desired pressures for all four segments as a function of time and transmits one setpoint per segment to the embedded nodes.
2. **Data logging:** assembles a complete data row (setpoints, sensor readback, and the latest mocap sample) and appends it to the log.
3. **Progress reporting (optional):** reports elapsed time and completion percentage.

Both the command and logging loops use drift correction (incrementing a scheduled wake time by 0.01 s each step) to maintain an approximately constant sampling interval over long runs.

Sequence automation

In addition to single-wave experiments, the controller supports an automated *sequence mode* that constructs a playlist of experiment items. Each item is parameterized by:

- the actuation waveform type (axial, circular, or triangular),
- the Segment 1 refill/set pressure (used for sensing-segment pre-inflation or refill actions), and
- the actuation pressure maximum $P_{\max}$ applied to Segments 2–4.

The playlist logic also inserts initialization holds (initial fill and prefill wait), cooldown periods between wave items, and (optionally) periodic refill pauses to re-pressurize Segment 1 during long trials.

On-disk data format

All trials are stored in a monthly HDF5 file (one file per month). Each experiment is saved as a group (e.g., `exp_011_sequence_Feb02_16h17m`) containing:

- a numeric dataset `data` with fixed column order,
- a string dataset `wave_types` containing the waveform label at each sample, and
- group attributes containing metadata (Arduino IDs, base pressures, mocap configuration, experiment duration, and a free-text description).

The numeric columns include: (i) time base (`step_id`, `time`); (ii) desired pressures `pd_{3,6,7,8}`; (iii) measured pressures `pm_{id}_{1..4}` for each node; (iv) mocap pose values `mocap_1_*`–`mocap_3_*` and `mocap_time_rel_s`; and (v) per-sample sequence parameters (`config_seg1_psi`, `config_max_psi`).

Post-processing and nomenclature

For downstream analysis, a cleaning stage maps abbreviated column names (e.g., `pd_3`, `pm_7_2`, `mocap_3_qx`) to descriptive names (e.g., `Desired_pressure_segment_1`, Measured_pressure_Segment_2, Rigid_body_3_qx) and drops unused sensor channels. The cleaned data can be exported to CSV with a consistent naming convention; the provided "Data Nomenclature" document summarizes one such CSV layout.

Chapter 2

PNEUMATIC SENSING TOPOLOGY STUDY FOR PHYSICAL RESERVOIR COMPUTING

Scope and motivation

Physical reservoir computing (PRC) uses the intrinsic dynamics of a physical system (here, a soft robot) as a computational resource. In a pneumatic soft arm, internal pressure signals can be used as a body-embedded state readout. A key design choice is the pneumatic *interconnection topology* among sensing cavities: sensing pouches can share air through a manifold ("coupled") or be sealed independently ("parallel"). This chapter reports a controlled experimental study comparing these two sensing-segment topologies on the same multi-segment arm.

The content and numerical results in this chapter match the accompanying MECC conference manuscript *Design of Pneumatic Soft Robots for Physical Reservoir Computing*.

System configurations

Figure 2.1 summarizes the PRC framework, the two sensing topologies, and the experimental factors.

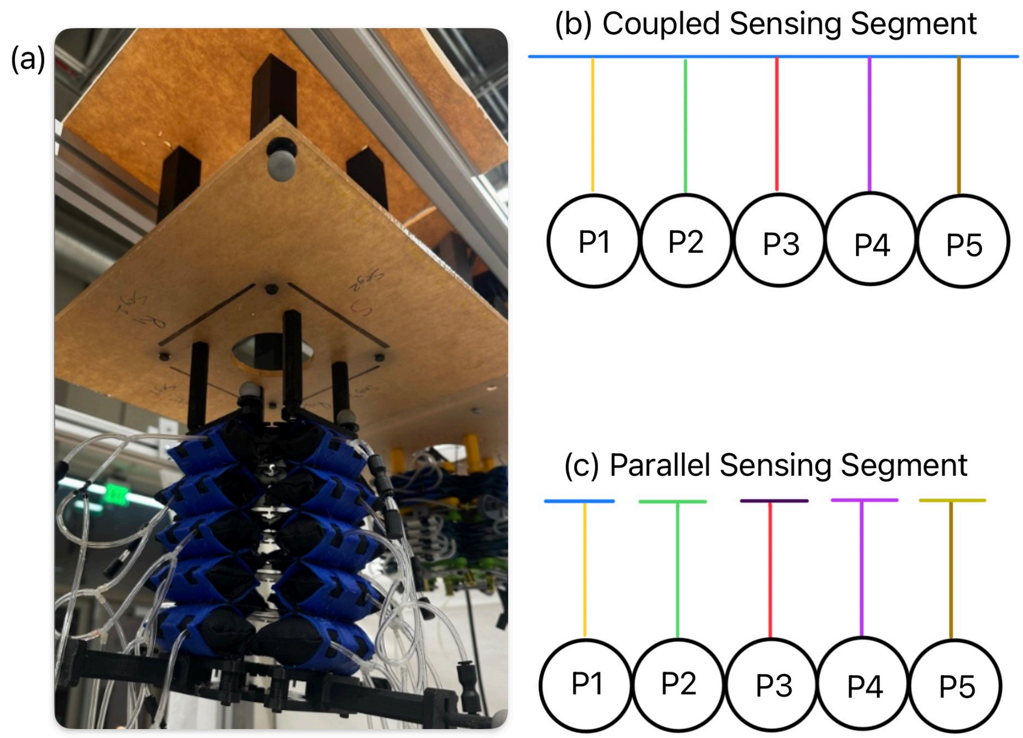

Figure 2.1: Experimental platform and pneumatic sensing topologies. (a) Fabric-based multi-segment pneumatic soft robotic arm with proximal sensing segment (Segment 1) containing five pressure-sensing pouches. (b) Coupled sensing topology: pouches P1-P5 are interconnected through a shared pneumatic manifold, allowing pressure equalization. (c) Parallel sensing topology: pouches P1-P5 are independently sealed, preventing air exchange and promoting signal independence.

Coupled topology

In the **coupled** configuration, the five sensing pouches of Segment 1 share a common manifold, allowing air exchange between pouches. Under deformation, pressure tends to redistribute through the shared pneumatic volume, which can reduce the independence of the five pressure signals.

Parallel topology

In the **parallel** configuration, each sensing pouch of Segment 1 is pre-inflated and then sealed. No air exchange occurs between pouches during operation. As a result, each pouch pressure responds primarily to local strain and contact, which can increase state diversity across the five pressure signals.

Experimental protocol

A total of 36 trials were collected: 18 trials in the coupled topology and 18 in the parallel topology. Each trial follows a factorial design across:

- three actuation waveforms (axial, circular, triangular) applied to Segments 2–4,
- three Segment 1 pre-inflation levels (1, 2, 3 PSI), and
- two actuation pressure maxima (5 and 10 PSI).

Each trial lasts approximately 200 s. To remove transients from initialization and shutdown, the first and last 10 s of each trial are discarded. For learning and evaluation, a chronological 70/30 train/test split is applied within each trial.

Open-loop actuation waveforms

The desired actuation pressure profiles for Segments 2–4 are summarized in Table 2.1. These waveforms are generated open-loop as time functions and streamed to the segment pressure regulators at $\approx 100\,\mathrm{Hz}$.

PRC formulation and evaluation

Reservoir readout and tapped-delay embedding

Let $\mathbf{u}(t) \in \mathbb{R}^3$ denote the commanded actuation pressure vector applied to Segments 2–4. Let $\mathbf{S}(t) = [s_1(t), \dots, s_5(t)]^\top \in \mathbb{R}^5$ denote the five Segment 1 pouch pressure signals.

Table 2.1: Desired actuation pressure profiles $p_{d,i}(t)$ applied to Segments 2–4. Here $p_{\max} \in \{5, 10\}$ PSI is the maximum actuation pressure, p_0 is the constant baseline used for non-actuated segments during *axial*, f=0.1 Hz (T=10 s), and $\tau = t \bmod T$. All commands are saturated to the regulator range $[0, p_{\max}]$.

Case	$p_{d,2}(t)$	$p_{d,3}(t)$	$p_{d,4}(t)$	pattern
hline Axial	p_0	$\frac{p_{\max}}{2}\left(1+\sin(2\pi f t)\right)$	p_0	—
Circular	$\frac{p_{\max}}{2}\left(1+\sin(2\pi f t)\right)$	$\frac{p_{\max}}{2}\left(1+\sin(2\pi f t+\frac{2\pi}{3})\right)$	$\frac{p_{\max}}{2}\left(1+\sin(2\pi f t+\frac{4\pi}{3})\right)$	○
Triangular	$\begin{cases} p_{\max}\left(1-\frac{\tau}{T/3}\right), & 0 \le \tau < \frac{T}{3} \\ 0, & \frac{T}{3} \le \tau < \frac{2T}{3} \\ p_{\max}\left(\frac{\tau-2T/3}{T/3}\right), & \frac{2T}{3} \le \tau < T \end{cases}$	$\begin{cases} p_{\max}\left(\frac{\tau}{T/3}\right), & 0 \le \tau < \frac{T}{3} \\ p_{\max}\left(1-\frac{\tau-T/3}{T/3}\right), & \frac{T}{3} \le \tau < \frac{2T}{3} \\ 0, & \frac{2T}{3} \le \tau < T \end{cases}$	$\begin{cases} 0, & 0 \le \tau < \frac{T}{3} \\ p_{\max}\left(\frac{\tau-T/3}{T/3}\right), & \frac{T}{3} \le \tau < \frac{2T}{3} \\ p_{\max}\left(1-\frac{\tau-2T/3}{T/3}\right), & \frac{2T}{3} \le \tau < T \end{cases}$	△

To provide short-term memory, the reservoir readout is embedded with a tapped delay line:

$$\mathbf{x}(t) = \left[\mathbf{S}(t),\, \mathbf{S}(t{-}1),\, \ldots,\, \mathbf{S}(t{-}n_\tau{+}1)\right] \in \mathbb{R}^{5n_\tau}, \tag{2.1}$$

with $n_\tau = 20$, giving 100 features (approximately 0.2 s of history at 100 Hz).

Bending-angle ground truth from motion capture

Let $\mathbf{q}_1(t)$ and $\mathbf{q}_3(t)$ be the unit quaternions of rigid bodies 1 and 3, and let $\otimes$ denote quaternion multiplication. The relative rotation is

$$\mathbf{q}_{\text{rel}}(t) = \mathbf{q}_1^{-1}(t) \otimes \mathbf{q}_3(t), \tag{2.2}$$

and the bending-angle magnitude (degrees) is computed from the scalar component $\mathbf{q}_{\text{rel},w}$ as

$$\theta(t) = \frac{180}{\pi}\, 2\cos^{-1}\!\left(|\mathbf{q}_{\text{rel},w}(t)|\right). \tag{2.3}$$

Prediction task and linear readout

A 1 s-ahead prediction task is evaluated by fitting a linear readout (Ridge regression, $\alpha = 0.01$) to predict $\theta(t+\Delta)$ from $\mathbf{x}(t)$ with $\Delta = 1\,\mathrm{s}$. Prediction accuracy is reported using normalized mean squared error (NMSE) and root mean squared error (RMSE).

In addition to the standard PRC readout that uses only $\mathbf{S}(t)$, an augmented "PRC+context" variant concatenates the known actuation command $\mathbf{u}(t)$ with the reservoir readout before forming the embedding.

Memory proxy: delay-decoding index

To probe the fading-memory property of the pneumatic reservoir, the delay-decoding index (DDI) is computed by measuring how well past actuation commands $\mathbf{u}(t-k)$ (for $k = 1$ to $K = 40$ steps) can be linearly decoded from the current reservoir state $\mathbf{S}(t)$. Larger DDI indicates that a broader range of past inputs remains decodable from the present reservoir readout.

State-diversity metrics

To quantify redundancy and effective dimensionality in the five pouch signals, the following metrics are computed per trial:

- mean inter-pouch Pearson correlation $\bar{r}$ (averaged across the 10 unique pouch pairs),
- the "unshared variance" proxy $1 - \bar{r}^2$,
- percent variance captured by the first principal component (PC1),
- participation ratio $\mathrm{PR} = (\sum_i \lambda_i)^2/(\sum_i \lambda_i^2)$ from PCA eigenvalues λ_i, and

- response diversity, defined as the coefficient of variation of per-pouch standard deviations.

Additional sensing-quality descriptors are reported (mean SNR, dynamic range, and a pressure–angle sensitivity proxy).

Results

Parallel topology improves PRC prediction accuracy

Figure 2.2 summarizes the 1 s-ahead prediction performance across all 36 trials.

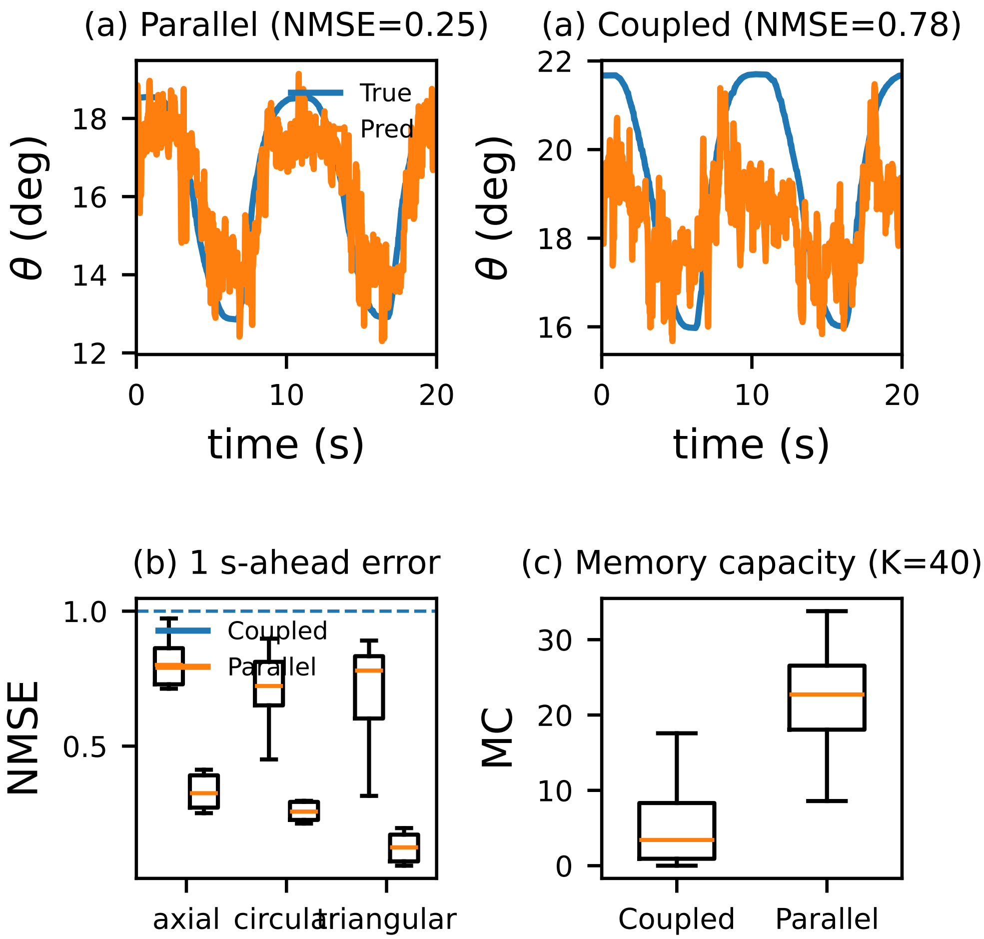

Figure 2.2: Performance comparison across waveforms and topologies. (a) Example prediction on an axial trial (parallel vs coupled). (b) NMSE distributions (18 trials per topology). (c) DDI distributions ($K = 40$).

Across all trials, the parallel topology yields substantially lower prediction error. The median NMSE decreases from 0.774 (coupled) to 0.242 (parallel), corresponding to a 69% reduction, with paired Wilcoxon signed-rank $p = 7.6 \times 10^{-6}$. Median RMSE decreases from 1.77° (coupled) to 0.93° (parallel). Parallel achieves NMSE$<$ 0.5 in 16/18 trials, compared with 2/18 in the coupled topology.

State diversity explains the performance gap

Figure 2.3 quantifies the redundancy in the five pouch signals.

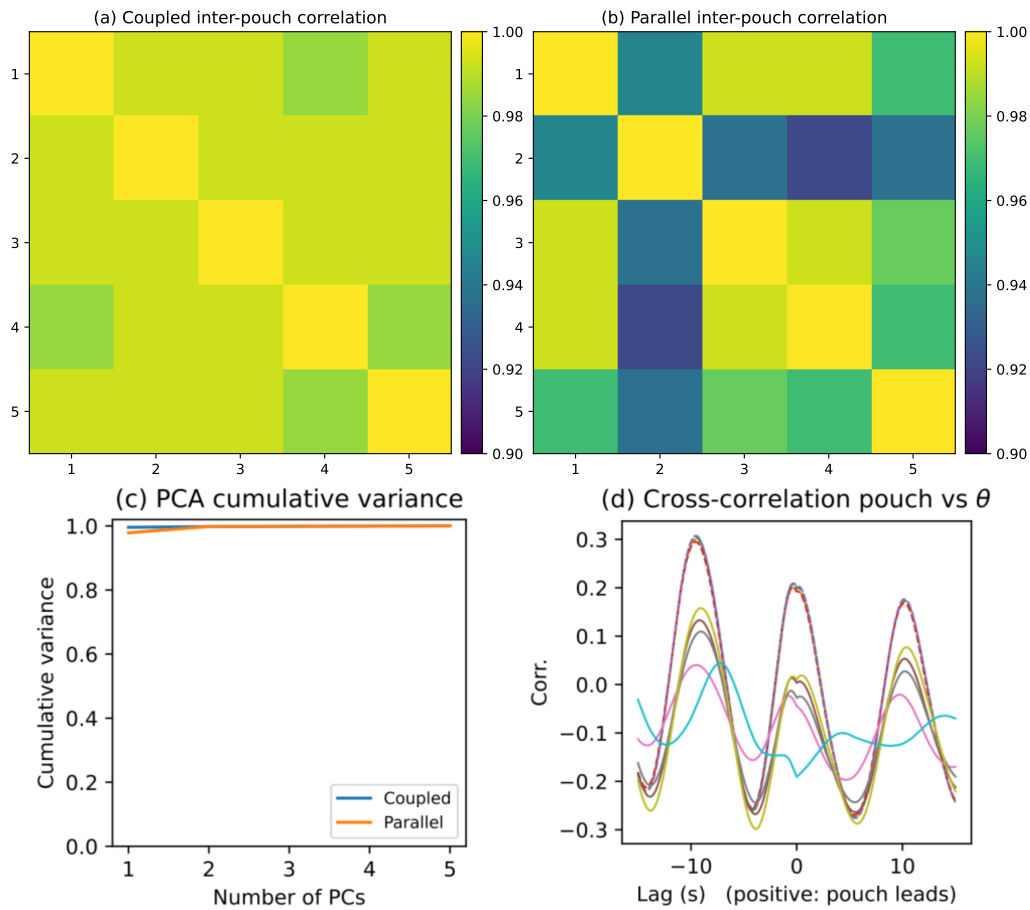

Figure 2.3: Signal diversity across trials. (a) Distribution of mean inter-pouch correlation $\bar{r}$. (b) Unshared variance proxy $1-\bar{r}^2$. (c) Relationship between redundancy and prediction error across trials.

The coupled manifold collapses the five pouch pressures into an effectively one-dimensional state. Across trials, coupled pouches exhibit mean inter-pouch correlation $\bar{r} = 0.992 \pm 0.005$ (median 0.993), with PC1 explaining $99.33 \pm 0.37\%$ of the variance and a participation ratio of 1.014 ± 0.008. In contrast, parallel pouches show lower redundancy ($\bar{r} = 0.931 \pm 0.055$, median 0.950), with PC1 explaining $95.09 \pm 3.68\%$ and a higher participation ratio (1.107 ± 0.086).

Although both $\bar{r}$ values are close to 1, small absolute changes near 1 correspond to large changes in non-overlapping information. Using $1-\bar{r}^2$ as a proxy, mean unshared variance increases from 0.0166 ± 0.0090 (coupled) to 0.129 ± 0.099 (parallel), i.e., approximately 8× more independent variance available to the readout.

Operating regime and pre-inflation effects

Figure 2.4 shows how bending variability and pre-inflation affect signal amplitude and prediction accuracy.

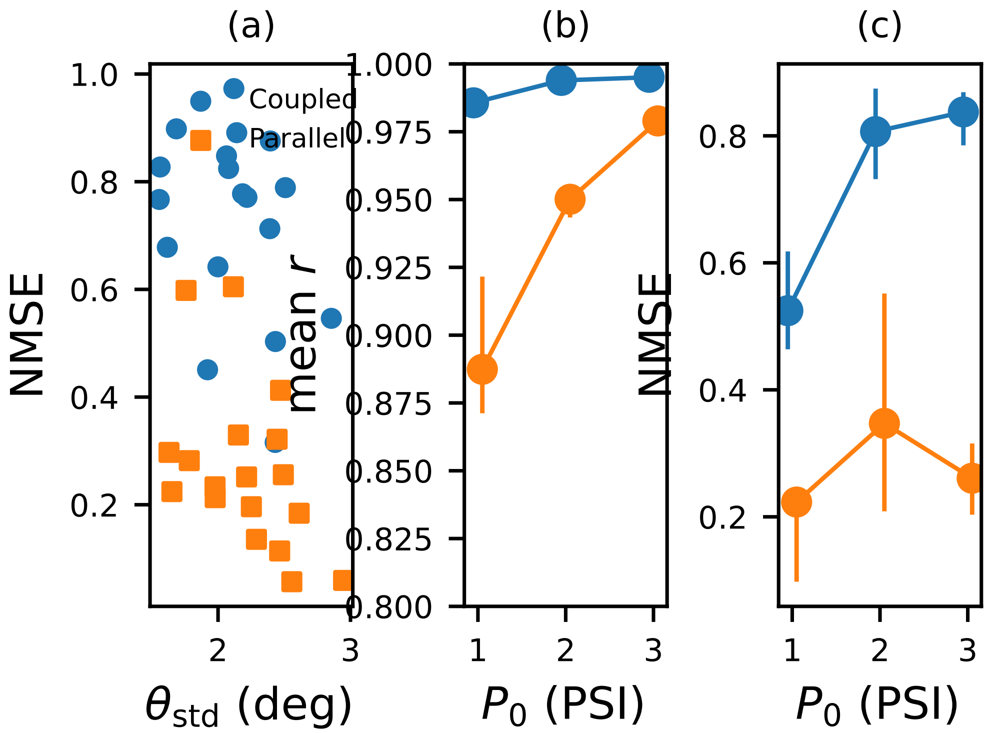

Figure 2.4: Operating regime and pre-inflation effects. (a) Bending variability vs NMSE. (b) Pre-inflation scales pouch signal amplitude. (c) Pre-inflation vs PRC accuracy.

Pre-inflation increases the amplitude of pouch pressure fluctuations for both topologies, but the parallel topology does not exhibit a monotonic improvement in NMSE with higher pre-inflation. In the coupled topology, higher pre-inflation degrades PRC accuracy and reduces DDI, consistent with further suppression of informative differential dynamics under shared-air coupling.

Figure 2.5 reports sensor ablations for the parallel topology.

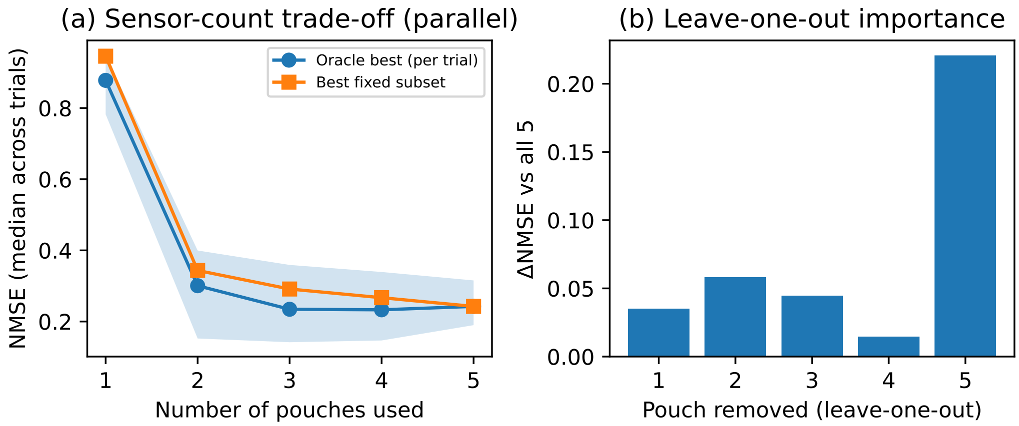

Figure 2.5: Sensor ablation study (parallel topology). (a) Median NMSE vs number of pouch sensors used (oracle and fixed subset selection). (b) Leave-one-out importance measured by NMSE change.

Performance improves rapidly as additional low-correlation pouches are added. The median NMSE of the best fixed 2-pouch subset is 0.344; the best fixed 3-pouch subset is 0.291; and using all five pouches yields 0.242. Leave-one-out analysis identifies pouch 5 as the most informative channel: removing it increases NMSE by a median of 0.221.

Summary and design implications

This controlled study isolates the effect of pneumatic interconnection topology on PRC performance for a fabric pneumatic arm. The results show that sealing sensing pouches (parallel topology) increases state diversity and substantially improves PRC prediction accuracy and memory proxies, even when signal amplitude and SNR are comparable. From a design perspective, topology is therefore a first-order parameter in pneumatic PRC: a small increase in inter-sensor independence can yield large gains in computational utility.

Chapter 3

OPEN-LOOP PRESSURE CONTROLLER AND FORWARD SIMULATION MODEL

Role of the controller in this thesis

The experimental platform is operated with an **open-loop, setpoint-based pressure controller**: the host computer generates desired pressure trajectories and streams them to the segment pressure regulators. No end-effector feedback is used to close the loop on pose; instead, repeatable excitation is achieved by commanding pressure setpoints at a fixed rate and recording the resulting pressures and motion.

This chapter provides two complementary components:

1. a technical description of the *open-loop pressure command generator* used to excite the arm (the "controller" used in experiments), and
2. a *forward simulation model* that maps commanded pressures to observed motion, enabling offline simulation of experimental outcomes.

Open-loop pressure command generation

Setpoint streaming and update rate

At each control step, the host transmits one scalar pressure setpoint to each segment (Segments 1–4). The command and logging loops are scheduled with drift correction at $\approx 100\,\mathrm{Hz}$ (0.01 s step), which matches the sampling rate of the recorded dataset.

Waveform primitives

Three waveform primitives are used to excite the actuation segments (Segments 2–4): **axial**, **circular**, and **triangular**. These match the definitions in Table 2.1.

Axial. A single actuation segment (Segment 3) is driven sinusoidally while the other two actuation segments are held at a constant baseline pressure p_0:

$$\begin{aligned} p_{d,2}(t) &= p_0, \\ p_{d,3}(t) &= \tfrac{p_{\max}}{2}\left(1 + \sin(2\pi f t)\right), \\ p_{d,4}(t) &= p_0, \end{aligned} \tag{3.1}$$

with $f = 0.1\,\mathrm{Hz}$ and saturation to $[0, p_{\max}]$.

Circular. Segments 2–4 are driven with three phase-shifted sinusoids (120° separation), producing a rotating pressure pattern:

$$p_{d,i}(t) = \tfrac{p_{\max}}{2}\left(1 + \sin(2\pi f t + \phi_i)\right), \quad \phi_i \in \left\{0, \tfrac{2\pi}{3}, \tfrac{4\pi}{3}\right\}. \tag{3.2}$$

Triangular. Segments 2–4 follow a piecewise-linear pattern over a cycle period $T = 10\,\mathrm{s}$, shifting pressure sequentially between segments (Table 2.1).

Initialization, cooldown, and Segment 1 handling

Segment 1 is used as a sensing segment in the PRC experiments and is treated differently from the actuation segments:

- **Initialization:** before actuation begins, the experiment sequence includes an initial fill and hold period to establish consistent starting conditions.
- **During actuation:** Segment 1 is commanded to 0 PSI during axial/circular/triangular waveforms so that its pouch pressures evolve passively under deformation.

- **Cooldown:** between waveform items, the controller enters a cooldown state that returns all segments to their baseline pressures.
- **Refill pauses (optional):** for long sequences, periodic pauses can be inserted to re-pressurize Segment 1 to a prescribed level.

Forward simulation via an identified FIR model

Objective

To simulate the measured motion resulting from an open-loop pressure program, we identify a discrete-time forward model that predicts kinematic outputs from the commanded segment pressures.

The simulation target is not closed-loop control. Instead, it provides:

- a compact surrogate model for reproducing observed trajectories, and
- a tool for testing alternative command profiles offline before running hardware experiments.

Model structure

Let $\mathbf{u}(t) = [p_{d,1}(t), p_{d,2}(t), p_{d,3}(t), p_{d,4}(t)]^\top \in \mathbb{R}^4$ denote the commanded pressures (PSI) and let $\mathbf{y}(t)$ denote a kinematic output.

A linear finite impulse response (FIR) model with N taps (memory length $N\Delta t$) is written as

$$\mathbf{y}(t) = \mathbf{b} + \sum_{k=0}^{N-1} \mathbf{H}_k \, \mathbf{u}(t-k) + \boldsymbol{\varepsilon}(t), \tag{3.3}$$

where $\mathbf{H}_k$ are constant gain matrices and $\boldsymbol{\varepsilon}(t)$ captures unmodeled dynamics.

Equation (3.3) is equivalent to linear regression on a tapped-delay feature vector built from $\mathbf{u}(t)$. The parameters $\{\mathbf{H}_k\}$ and $\mathbf{b}$ are identified by Ridge regression ($\alpha = 0.01$).

Outputs and preprocessing

Two outputs are simulated:

- **Bending angle** $\theta(t)$ computed from mocap quaternions using Eq. (2.3), and
- **Tip position** $(x(t), z(t))$ from the distal rigid body position (meters).

To remove transient effects, the first and last 10 s of each waveform segment are discarded. Within each trial, a chronological 70/30 train/test split is used, and performance is reported on the held-out test segment.

Representative simulation results

Table 3.1 summarizes FIR simulation accuracy for three representative trials (one per waveform type) from the February dataset. The model uses $N = 30$ taps (0.30 s of command history at 100 Hz).

Table 3.1: FIR forward-model simulation accuracy on three representative trials (test segment only). RMSE is reported for bending angle (degrees) and tip position (meters).

Trial (waveform)	RMSE_θ	NMSE_θ	RMSE_x	NMSE_x	RMSE_z	NMSE_z
Axial (`exp_007`)	0.413°	0.045	3.08×10^{-4}	0.359	1.08×10^{-3}	0.050
Circular (`exp_011`)	0.413°	0.052	9.57×10^{-4}	0.027	1.11×10^{-3}	0.045
Triangular (`exp_017`)	0.538°	0.059	1.57×10^{-3}	0.045	1.48×10^{-3}	0.051

Figure 3.1 shows measured versus simulated bending angle and tip trajectory for the circular trial `exp_011`. Figure 3.2 shows the corresponding open-loop pressure commands for Segments 2–4.

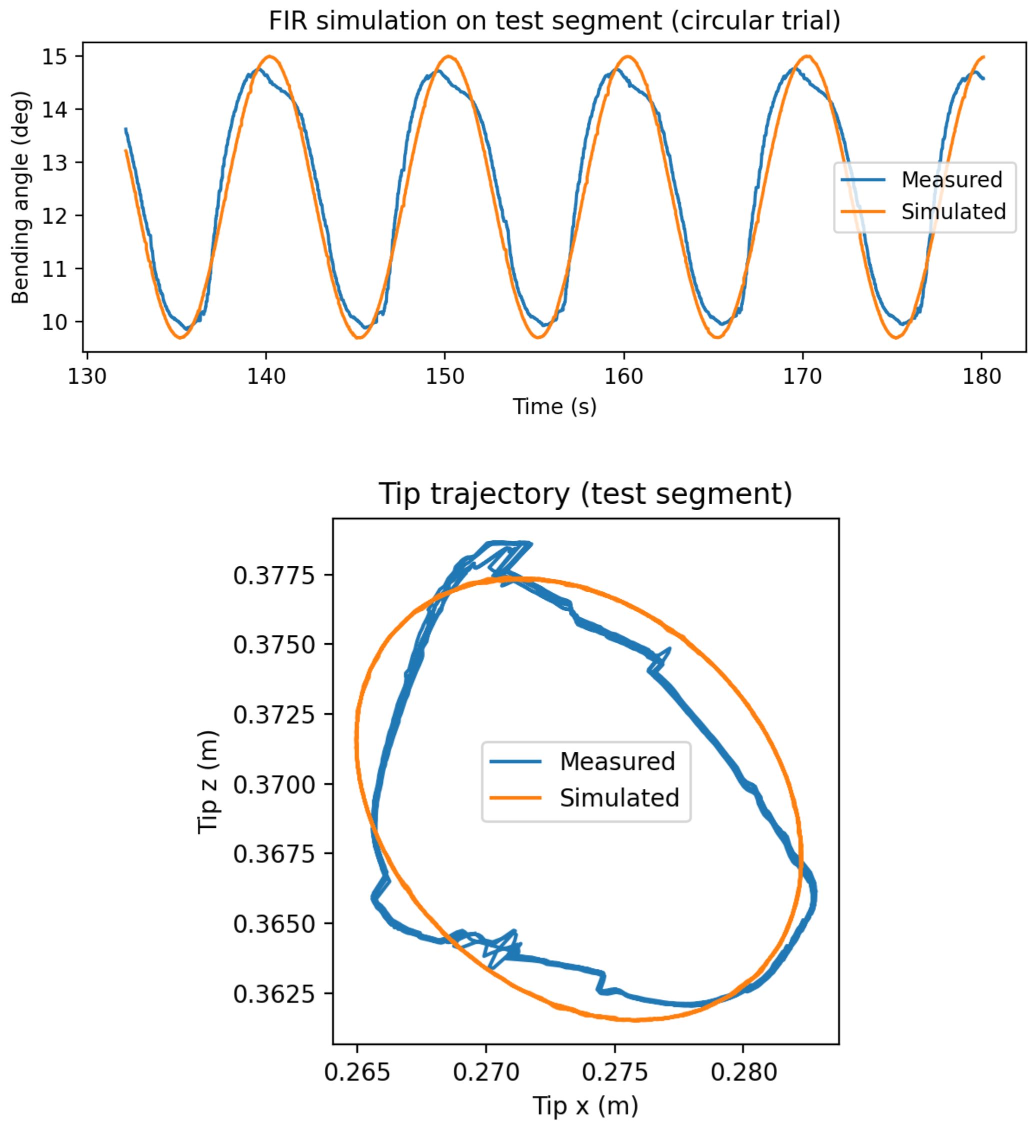

Figure 3.1: Forward simulation on the circular trial `exp_011`. Top: measured vs simulated bending angle on the held-out test segment. Bottom: measured vs simulated tip trajectory in the x–z plane (test segment).

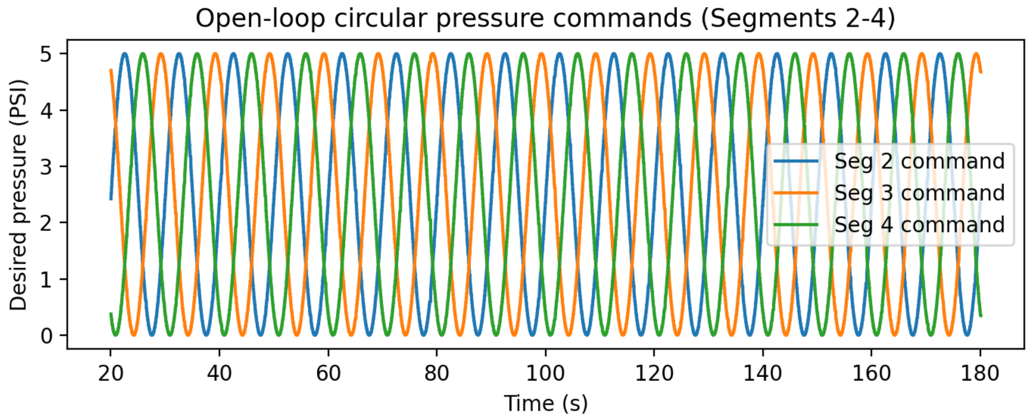

Figure 3.2: Open-loop circular pressure commands applied to Segments 2–4 in trial `exp_011` (after trimming initial/final 10 s).

Limitations and use

The FIR simulator in Eq. (3.3) is intentionally simple and is intended as a pragmatic surrogate model for offline reproduction of observed trajectories under similar command profiles. Because it is identified from data within each trial (chronological train/test split), its primary use is to simulate outcomes for command profiles that remain close to the training distribution (same waveform family and operating regime). Extending simulation across trials, across topologies, or across substantially different pressure programs would require either (i) pooled identification across a wider dataset, or (ii) a nonlinear/state-space model.

APPENDIX A

RAW DATA EXAMPLES

This appendix provides representative **raw data plots** from the February 2026 dataset stored in `2026_February.h5`. For each actuation waveform (axial, circular, triangular), the figures show:

- the open-loop pressure setpoints streamed to the embedded nodes (desired pressures),
- the corresponding pressure tracking behavior (desired vs measured) for the actuation segments (Segments 2–4),
- the five Segment 1 pouch-pressure channels recorded during actuation (the PRC reservoir readout), and
- mocap-derived kinematics (bending angle magnitude θ and distal rigid-body trajectory in the x–z plane).

Unless otherwise stated, time-series plots are shown over the **trimmed waveform interval**: the waveform-specific segment of each trial (`wave_types` = axial/circular/triangular) with the first and last 10 s removed, matching the preprocessing used in Chapters 2–3.

Logger timing and motion-capture sample age

appendix_figures/A14_timing_mocap_age_exp011.png

Figure A.1: Raw timing statistics for experiment `exp_011_sequence_Feb02_16h17m`: per-sample logging interval dt (top) and its histogram (middle), and age of the most recent mocap sample at each logging instant computed as (`time` − `mocap_time_rel_s`) (bottom).

Axial waveform raw data (`exp_007_sequence_Feb02_15h22m`)

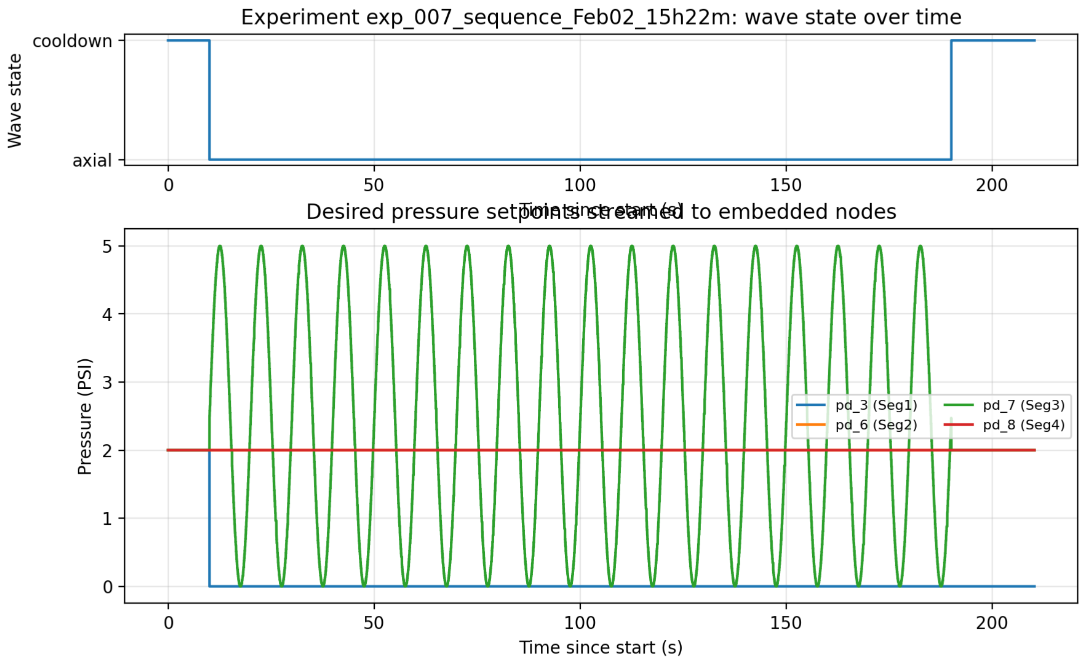

Figure A.2: Raw desired pressure setpoints streamed to the embedded nodes for the axial trial `exp_007_sequence_Feb02_15h22m`. The top panel indicates the waveform state (cooldown vs axial), and the bottom panel shows the four commanded segment pressures (`pd_3`, `pd_6`, `pd_7`, `pd_8`).

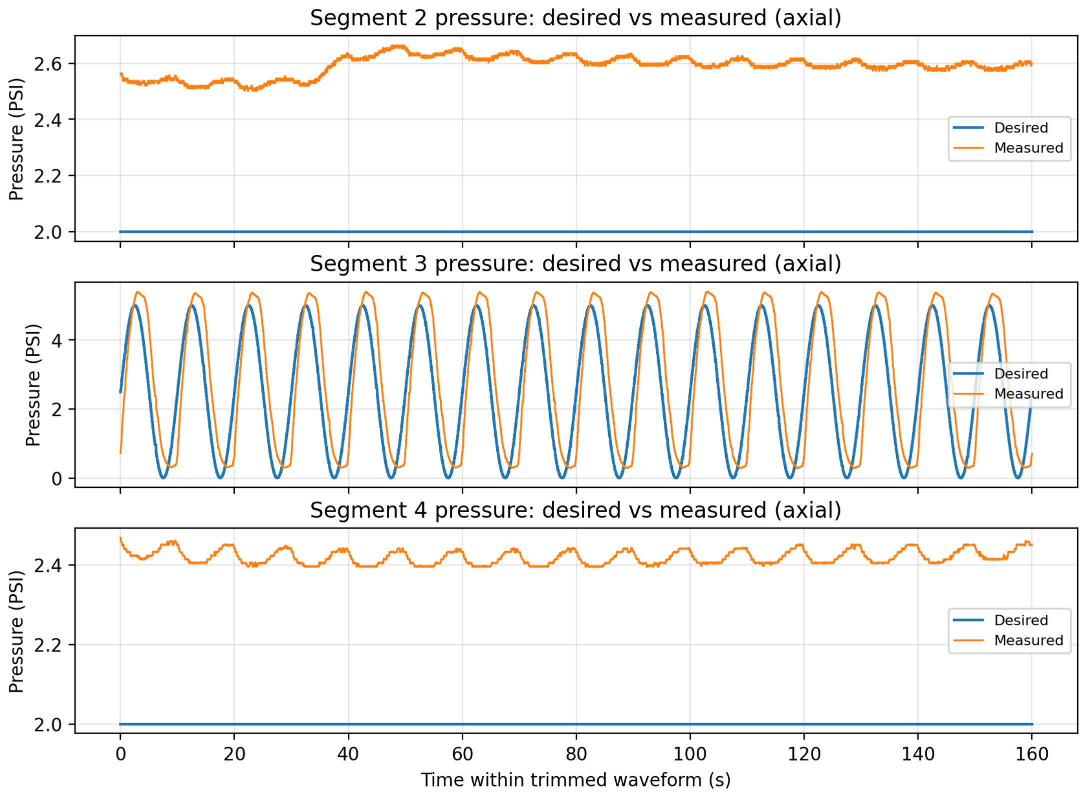

Figure A.3: Raw pressure tracking for the axial waveform segment: desired vs measured pressures for Segments 2–4 in the trimmed waveform interval.

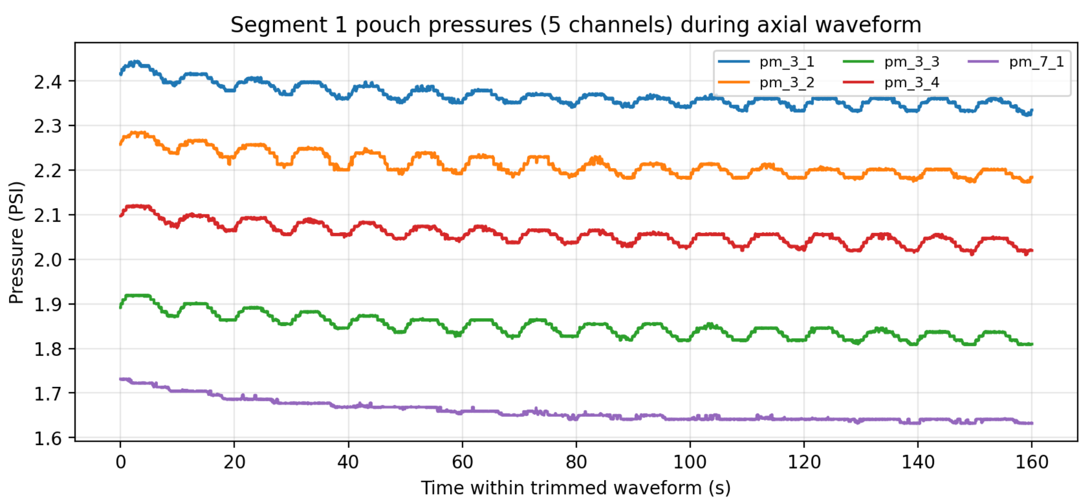

Figure A.4: Raw Segment 1 pouch-pressure channels (five channels) during the trimmed axial waveform interval. These channels are used as the PRC reservoir readout in Chapter 2.

appendix_figures/A11_mocap_theta_tip_exp007_axial.png

Figure A.5: Raw mocap-derived kinematics for the trimmed axial waveform interval of `exp_007_sequence_Feb02_15h22m`. Left: bending angle magnitude θ computed from the relative rotation between rigid bodies 1 and 3 (Eq. 2.3). Right: distal rigid-body trajectory in the x–z plane.

Circular waveform raw data (`exp_011_sequence_Feb02_16h17m`)

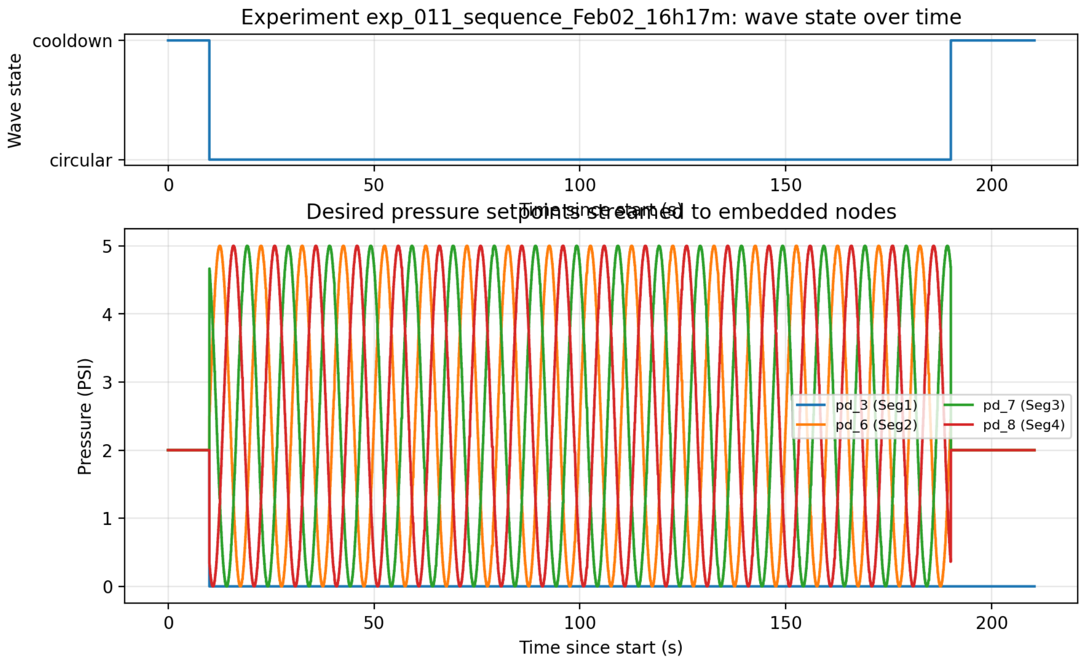

Figure A.6: Raw desired pressure setpoints streamed to the embedded nodes for the circular trial `exp_011_sequence_Feb02_16h17m`. The circular waveform drives Segments 2–4 with phase-shifted sinusoids, while Segment 1 is held at 0 PSI during the actuation segment.

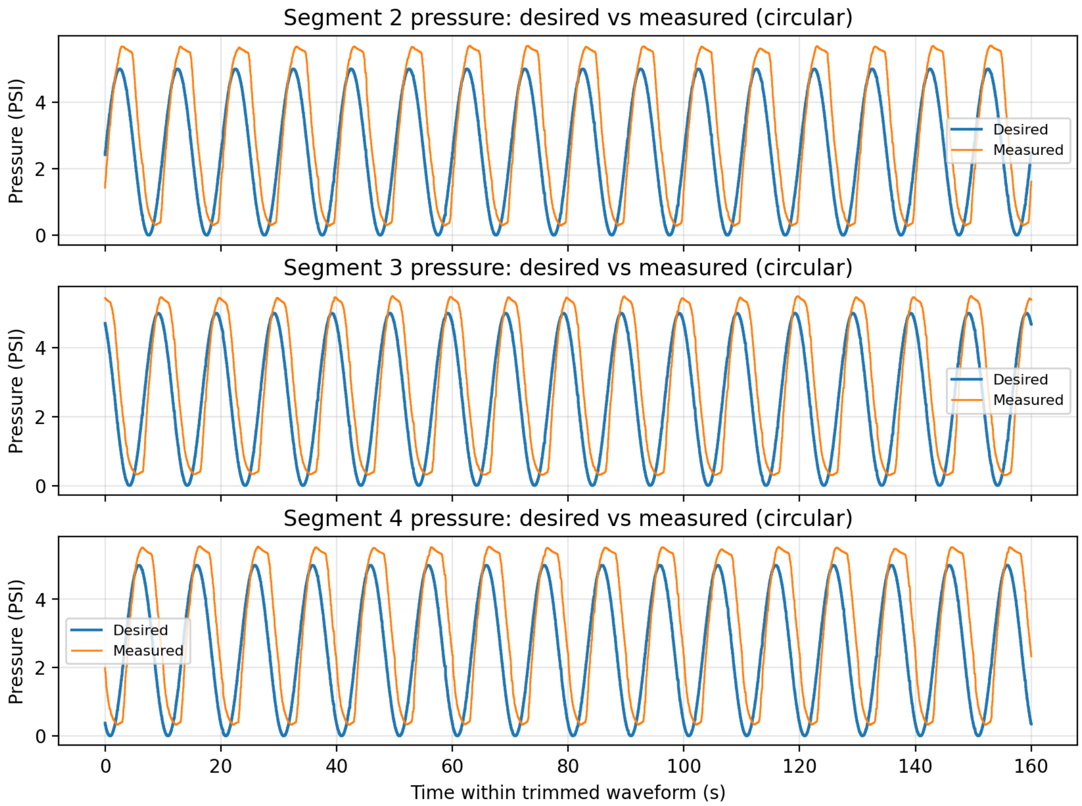

Figure A.7: Raw pressure tracking for the circular waveform segment: desired vs measured pressures for Segments 2–4 in the trimmed waveform interval.

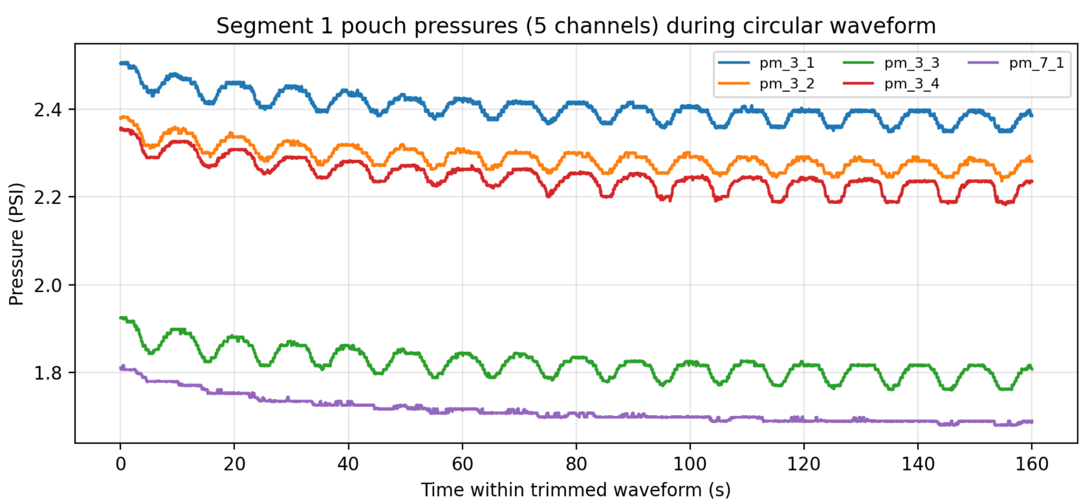

Figure A.8: Raw Segment 1 pouch-pressure channels (five channels) during the trimmed circular waveform interval.

appendix_figures/A12_mocap_theta_tip_exp011_circular.png

Figure A.9: Raw mocap-derived kinematics for the trimmed circular waveform interval of `exp_011_sequence_Feb02_16h17m`. Left: bending angle magnitude θ (Eq. 2.3). Right: distal rigid-body trajectory in the x–z plane.

Triangular waveform raw data (`exp_017_sequence_Feb02_16h50m`)

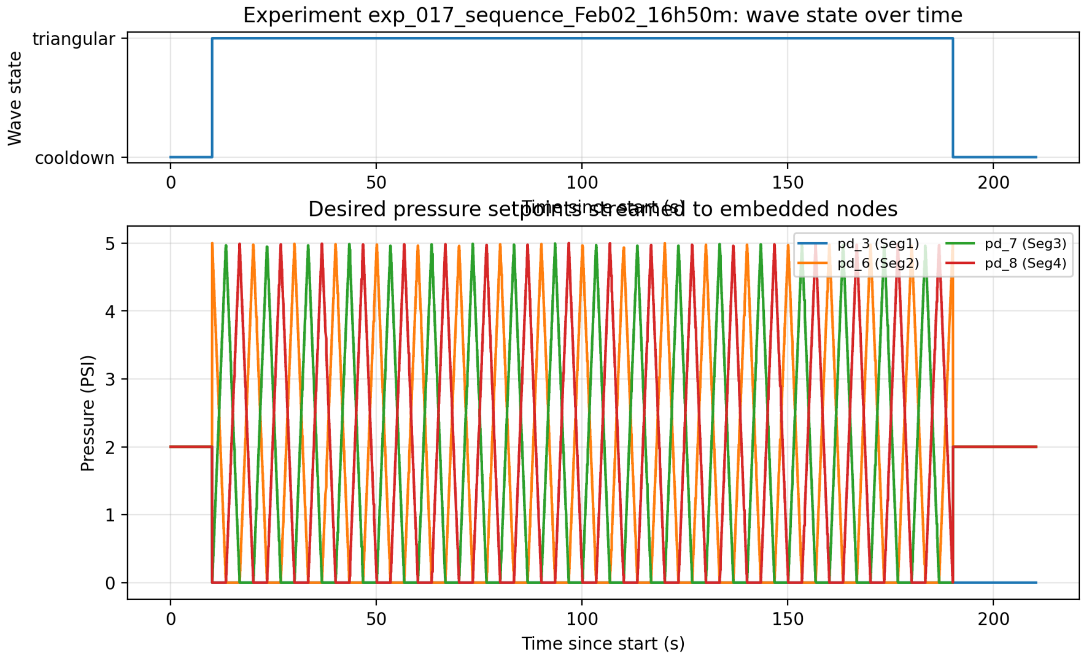

Figure A.10: Raw desired pressure setpoints streamed to the embedded nodes for the triangular trial `exp_017_sequence_Feb02_16h50m`. The top panel indicates the waveform state (cooldown vs triangular), and the bottom panel shows the commanded segment pressures.

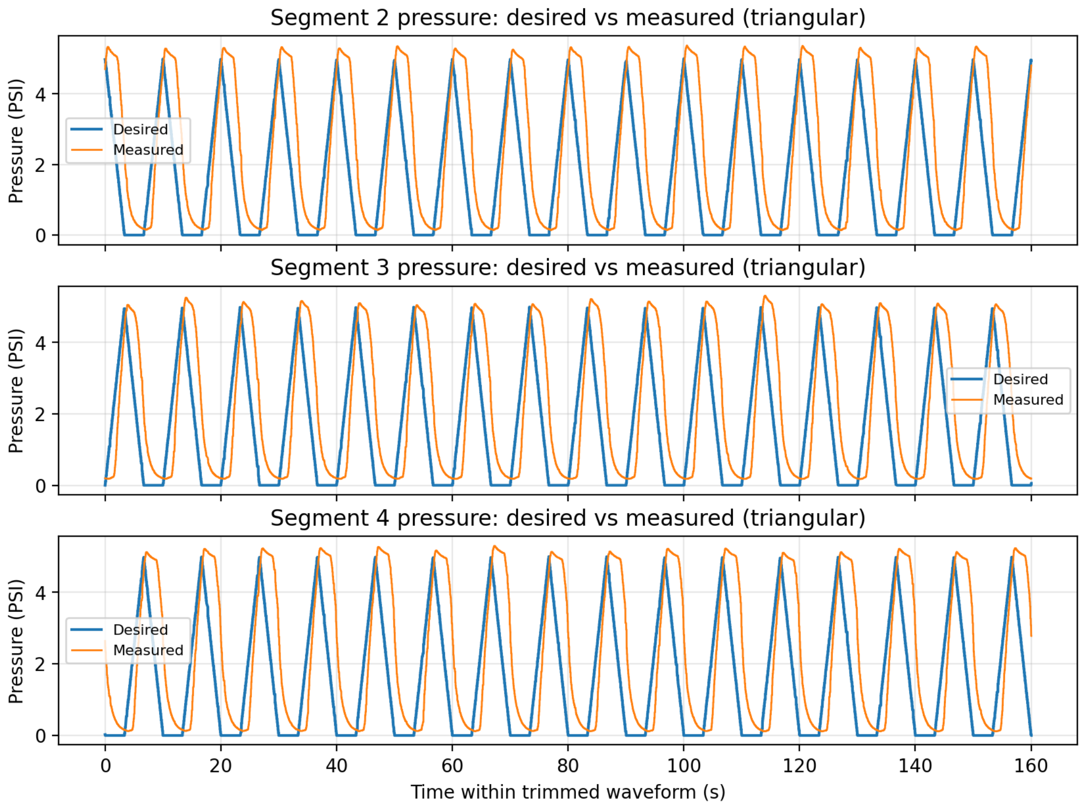

Figure A.11: Raw pressure tracking for the triangular waveform segment: desired vs measured pressures for Segments 2–4 in the trimmed waveform interval.

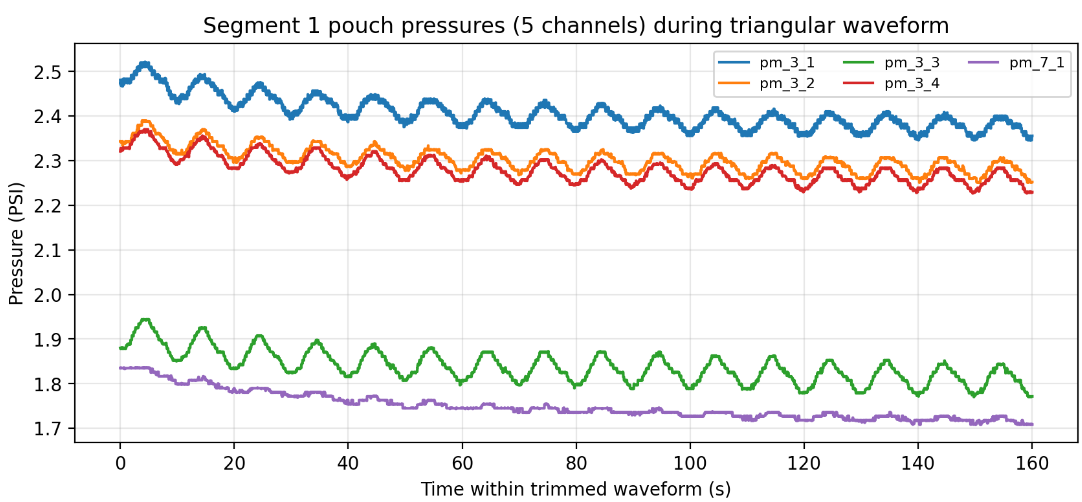

Figure A.12: Raw Segment 1 pouch-pressure channels (five channels) during the trimmed triangular waveform interval.

appendix_figures/A13_mocap_theta_tip_exp017_triangular.png

Figure A.13: Raw mocap-derived kinematics for the trimmed triangular waveform interval of `exp_017_sequence_Feb02_16h50m`. Left: bending angle magnitude θ (Eq. 2.3). Right: distal rigid-body trajectory in the x–z plane.